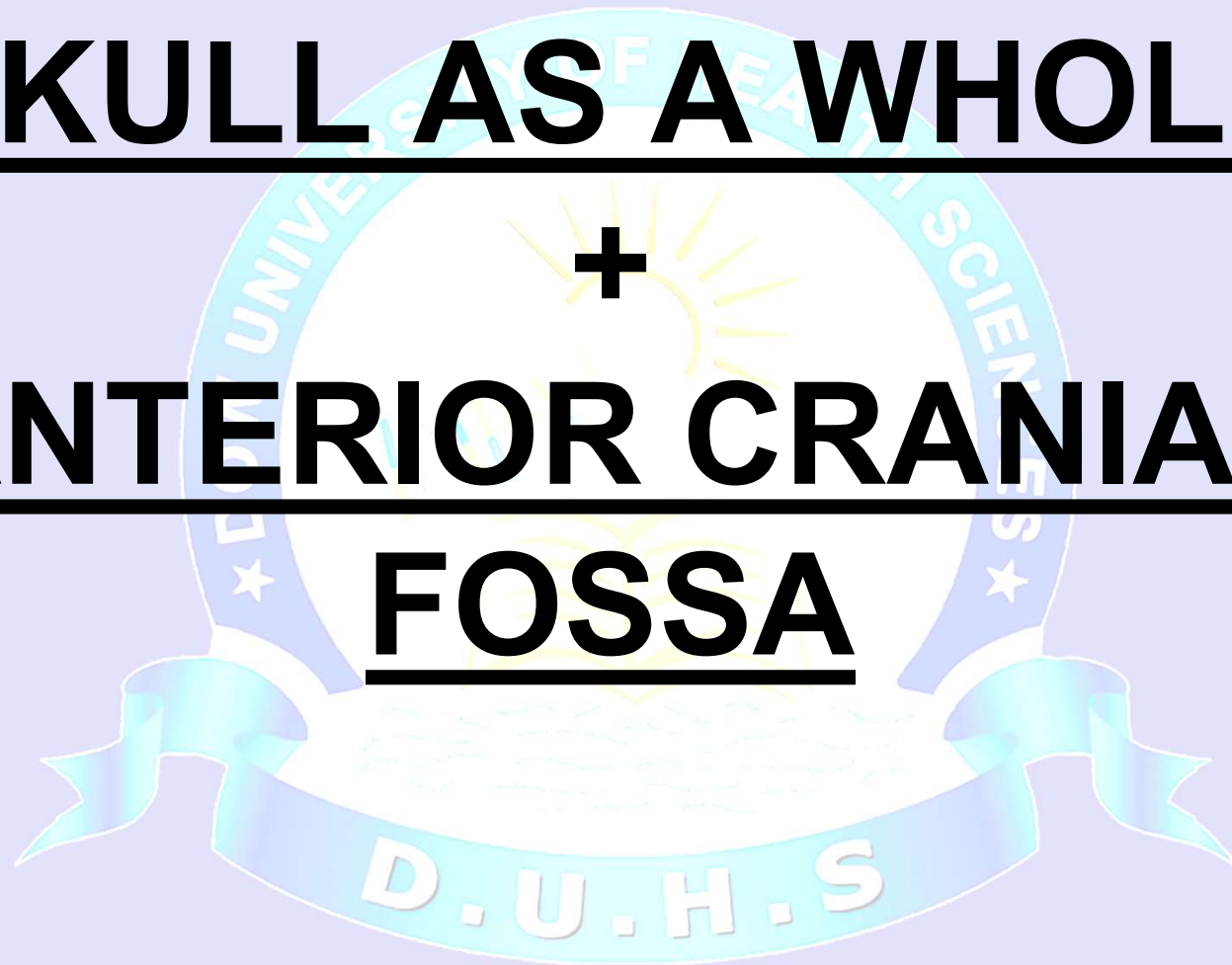


SKULL AS A WHOLE

+

ANTERIOR CRANIAL

FOSSA



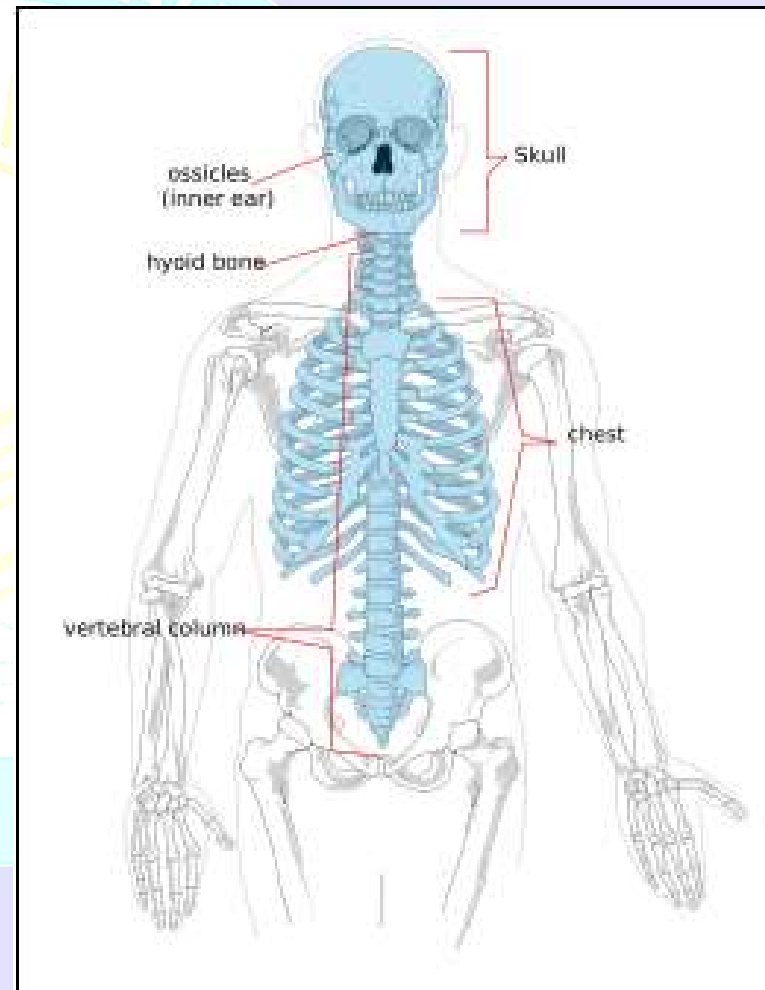
LEARNING OBJECTIVES

At the end of this lecture , the student should be able to know:

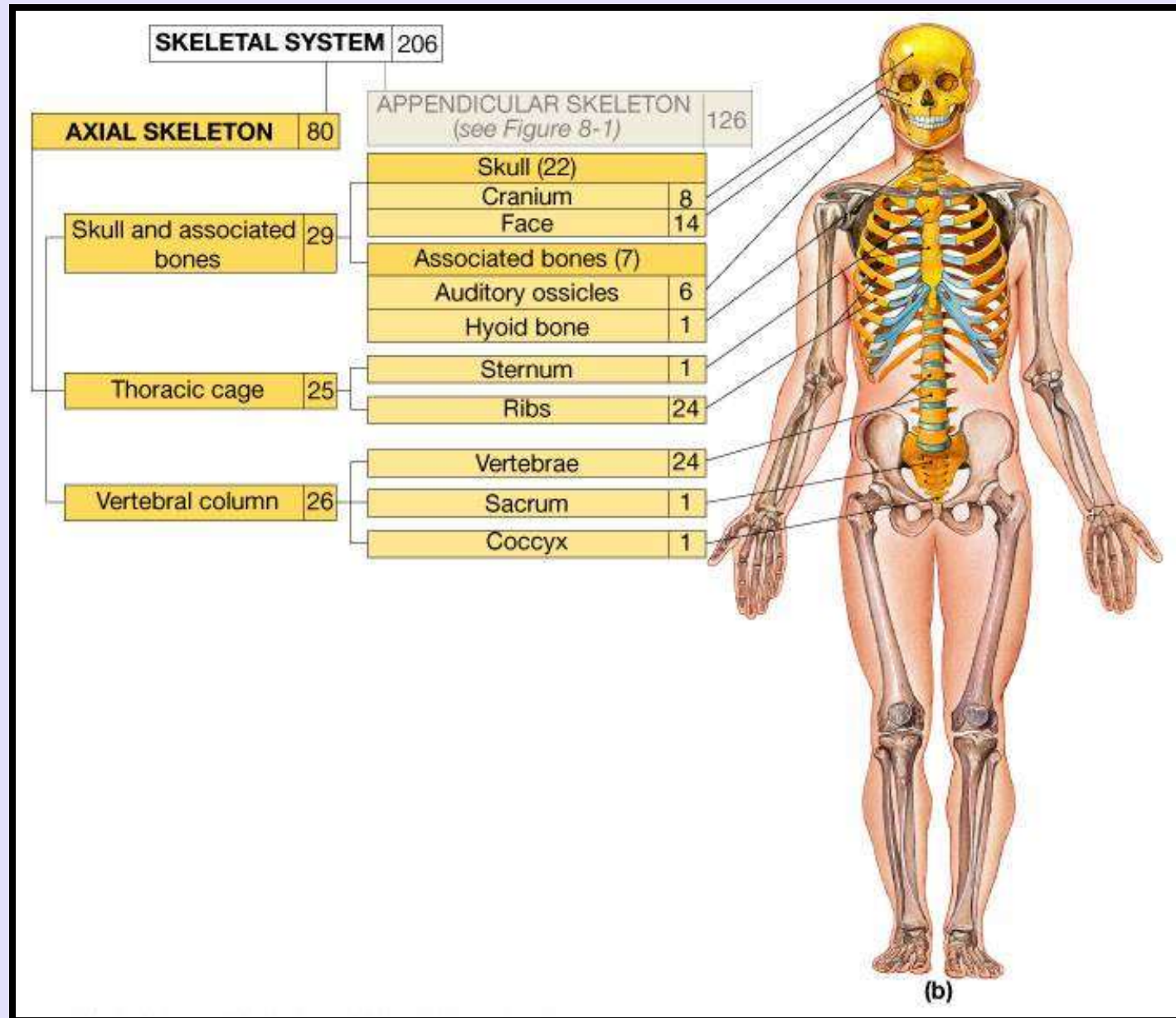
- Parts of skeleton (axial and appendicular)
- Parts of skull
- Sutures of skull
- Different bones of skull
- Different views (Norma) of skull
- Interior of the skull
- Divisions of the cranial fossa
- Anterior cranial fossa
- Foramens and structures passing through them

The Axial Skeleton

- Eighty (80) bones
- It is composed of five parts;
 - Skull
 - Ossicles of the middle ear
 - Hyoid bone of the throat
 - Vertebral column
 - Bony thorax



The Axial Skeleton

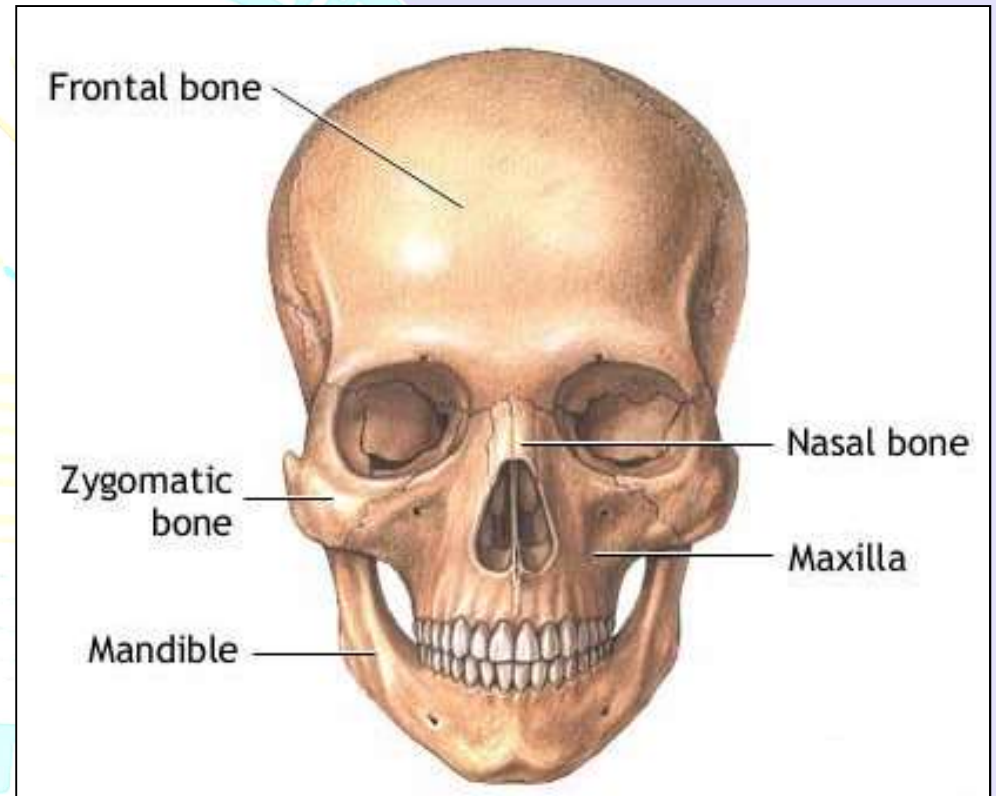


The skull

- The skull, body's most complex bony structure, is formed by the cranium and facial bones (bones of face)
- **Cranium** –
 - cranium encloses cranial cavity
 - protects the brain and is the site of attachment for head and neck muscles
- **Facial bones** –
 - Supply the framework of the face, the sense organs, and the teeth
 - Provide ,surround and protect the entrances to the respiratory and digestive tracts
 - Anchor the facial muscles of expression

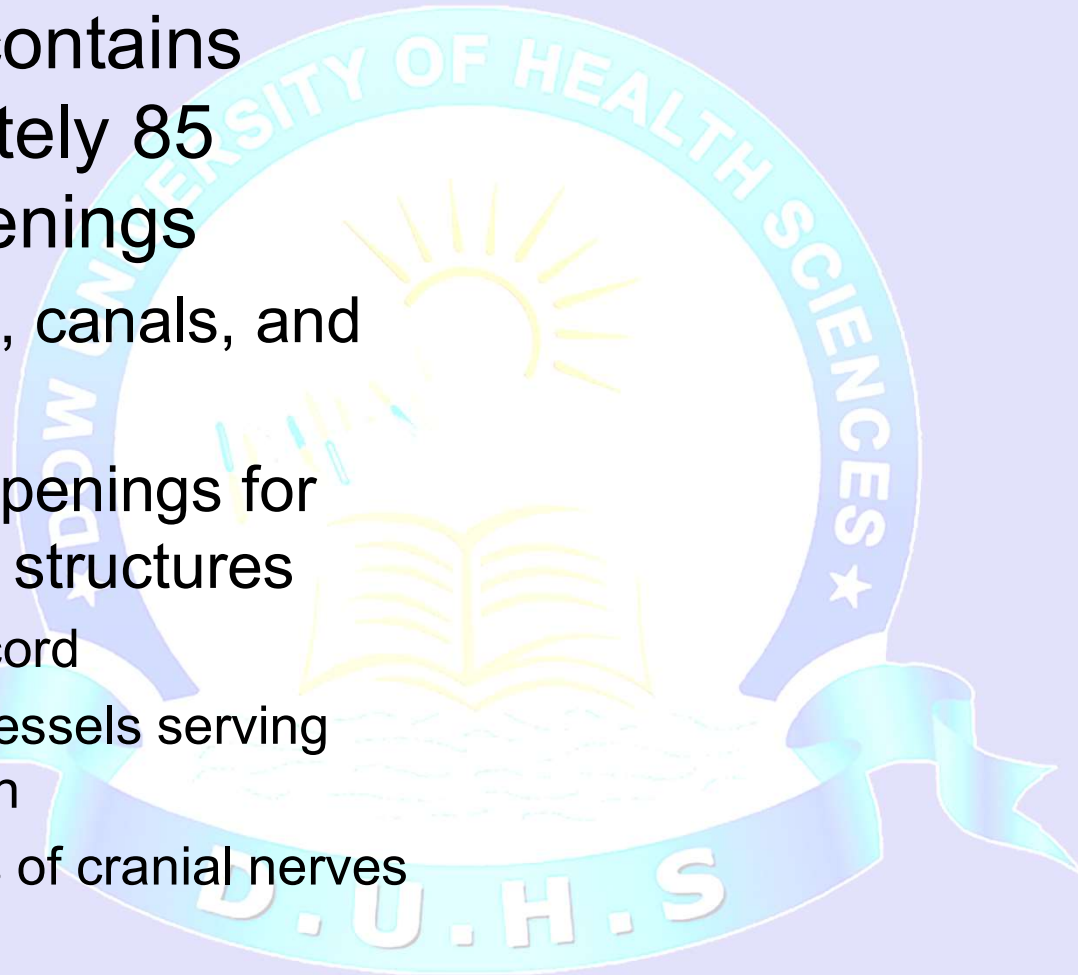
Bones of skull

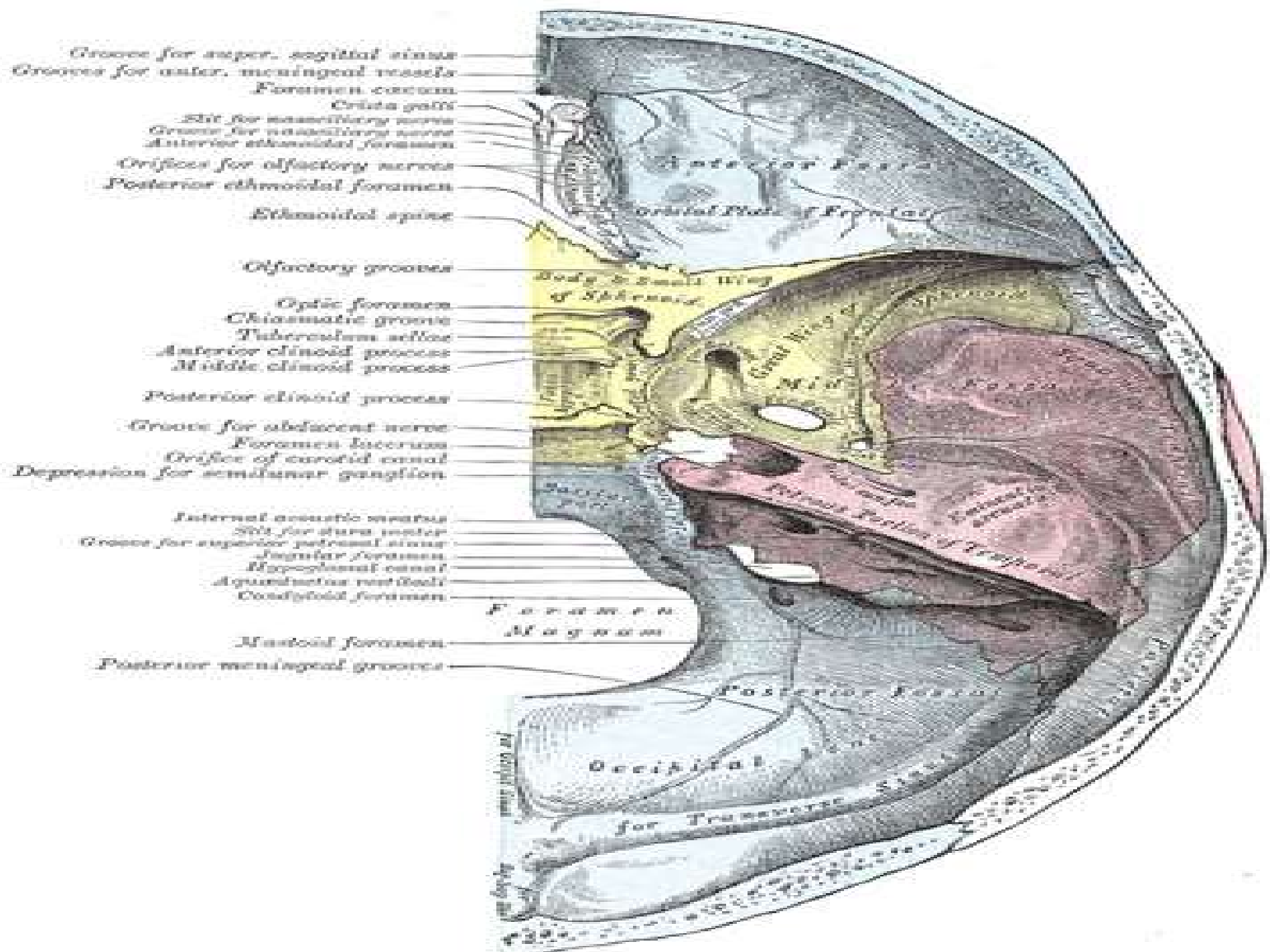
- Formed from eight large bones
 - Paired bones include
 - Temporal bones
 - Parietal bones
 - Unpaired bones include
 - Frontal bone
 - Occipital bone
 - Sphenoid bone
 - Ethmoid bone



Overview of Skull Geography

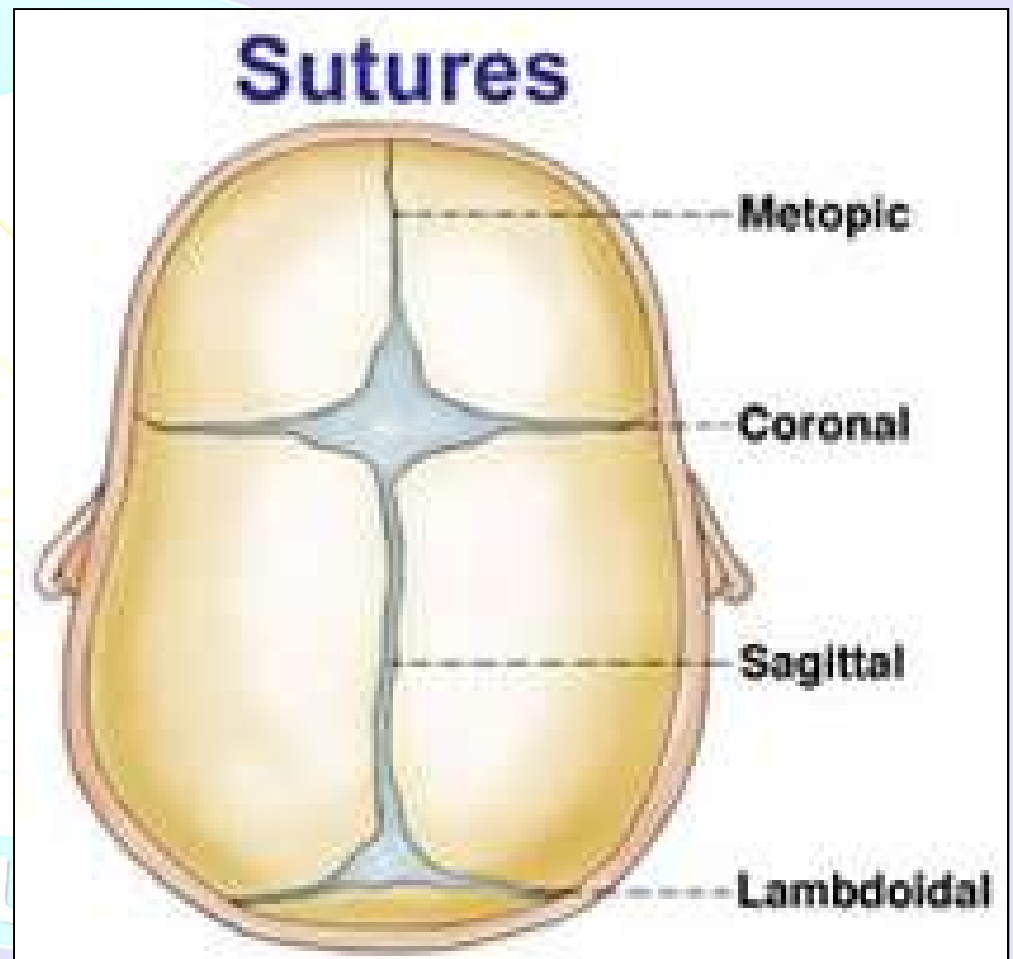
- The skull contains approximately 85 named openings
 - Foramina, canals, and fissures
 - Provide openings for important structures
 - Spinal cord
 - Blood vessels serving the brain
 - 12 pairs of cranial nerves



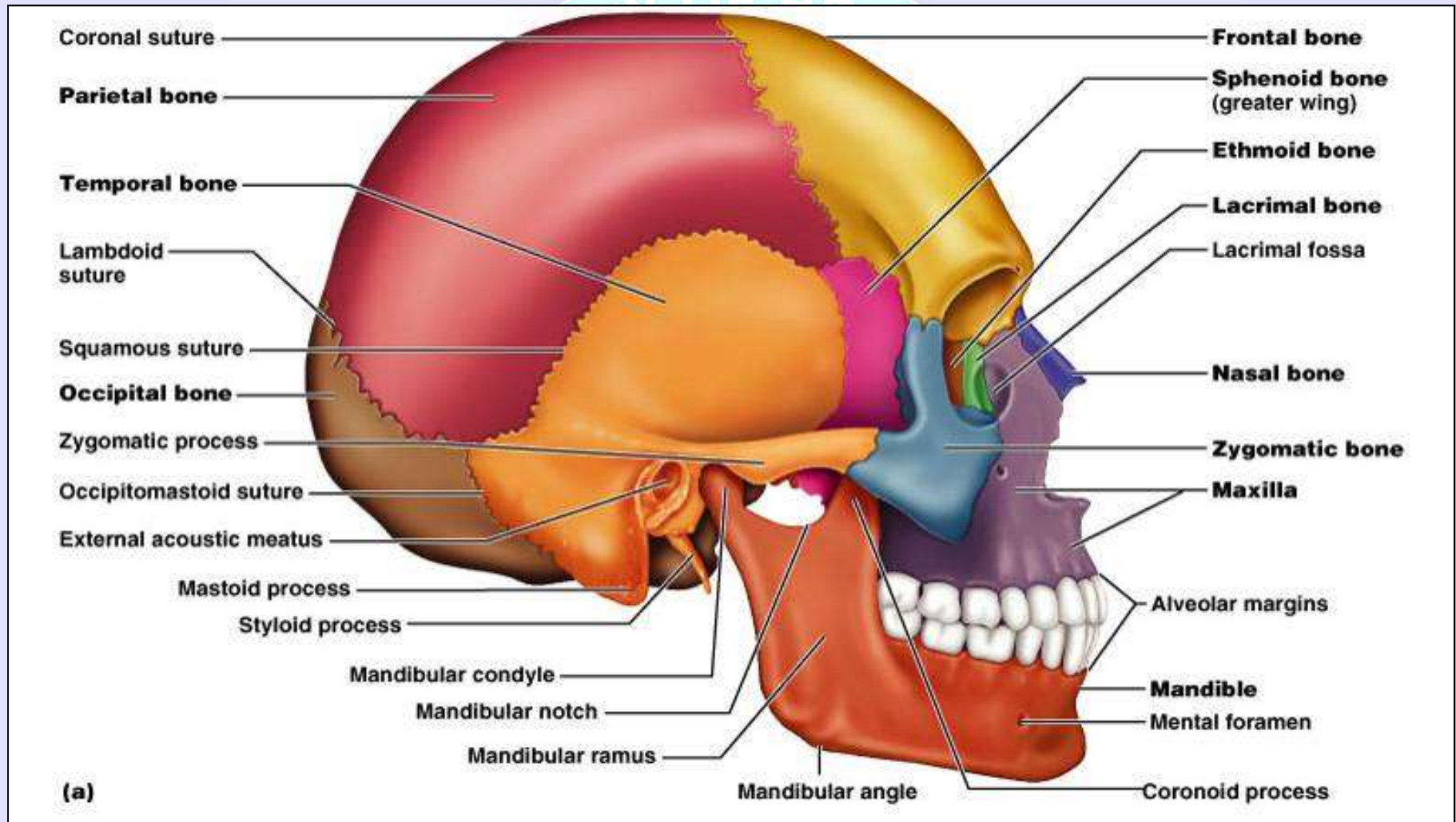


SUTURES

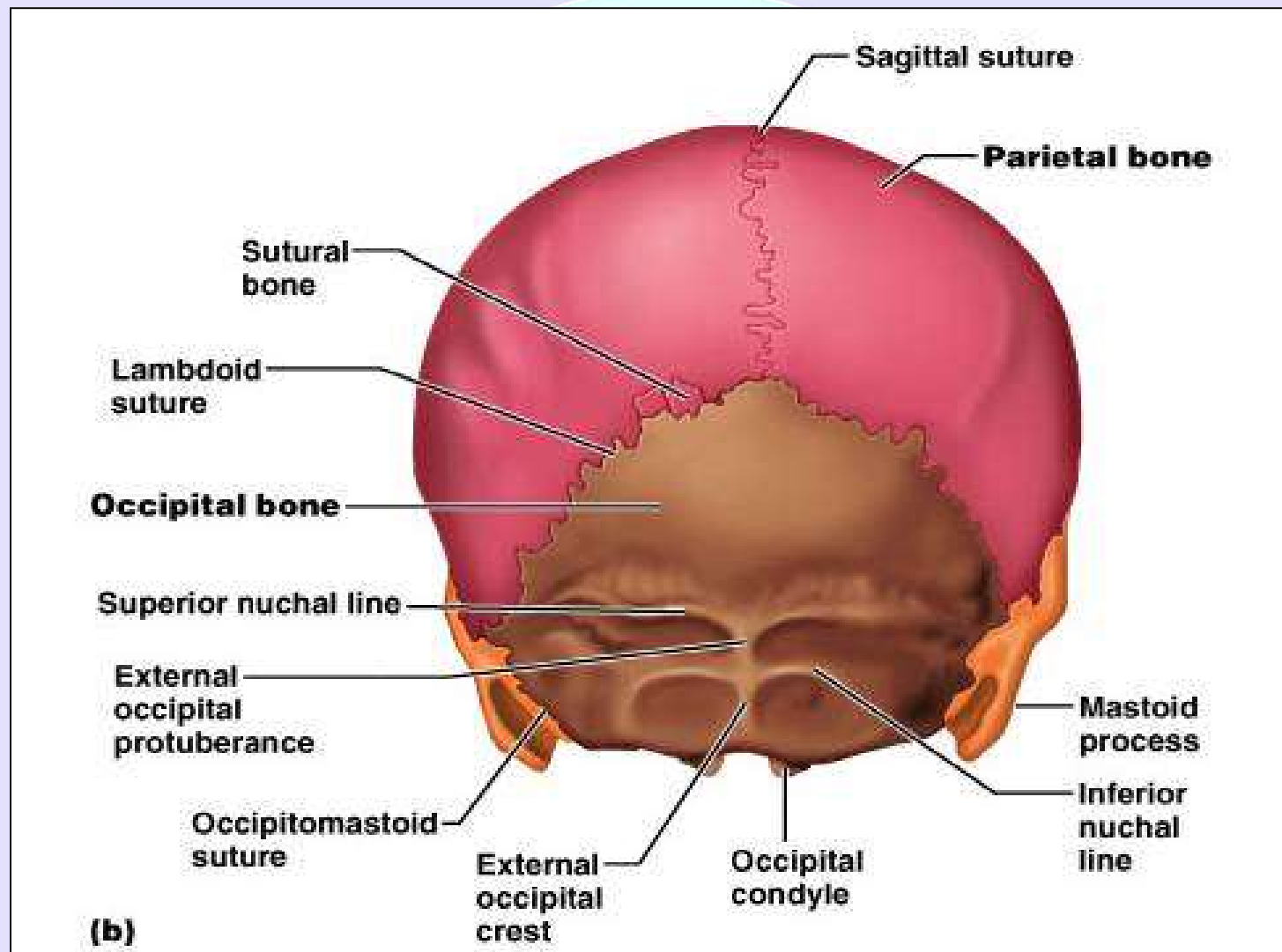
- A **suture** is a type of fibrous joint which only occurs in the skull (or "cranium").
- A tiny amount of movement is permitted at sutures, which contributes to the compliance and elasticity of the skull.
- Main sutures are:
 - Sagittal
 - Coronal or Frontal
 - Squamosal
 - Lambdoidal
 - Sutural (Wormian) Bones
 - Usually found in Lambdoidal suture



SUTURES OF LATERAL SIDE

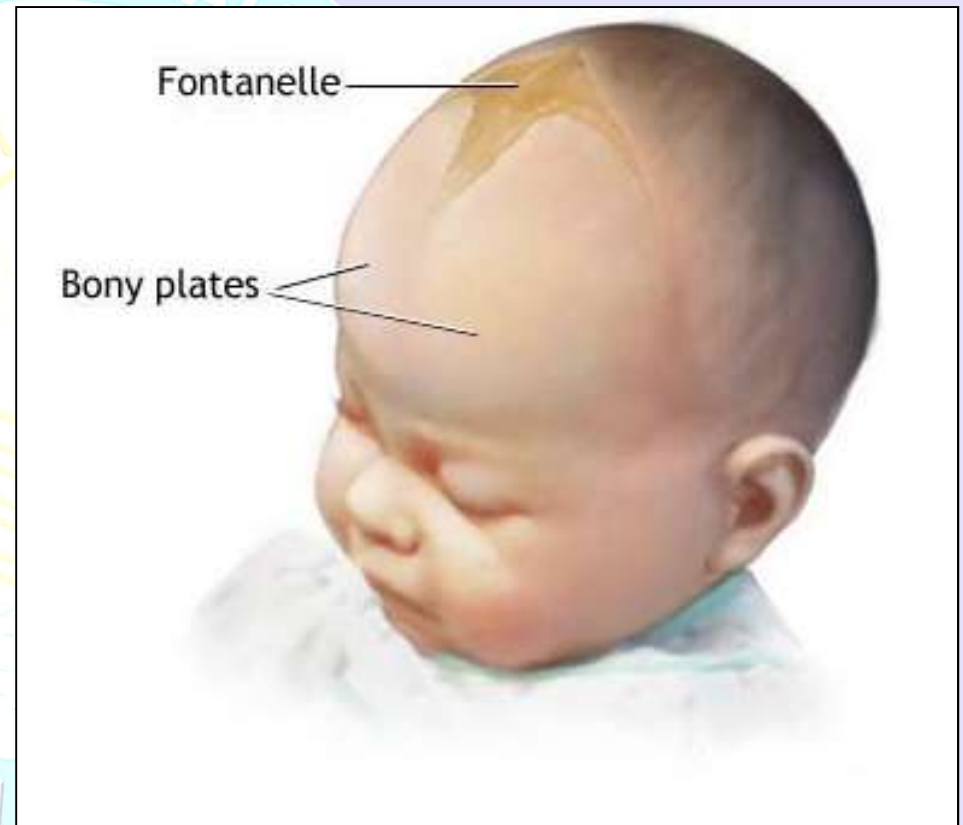


SUTURES OF OCCIPITAL REGION

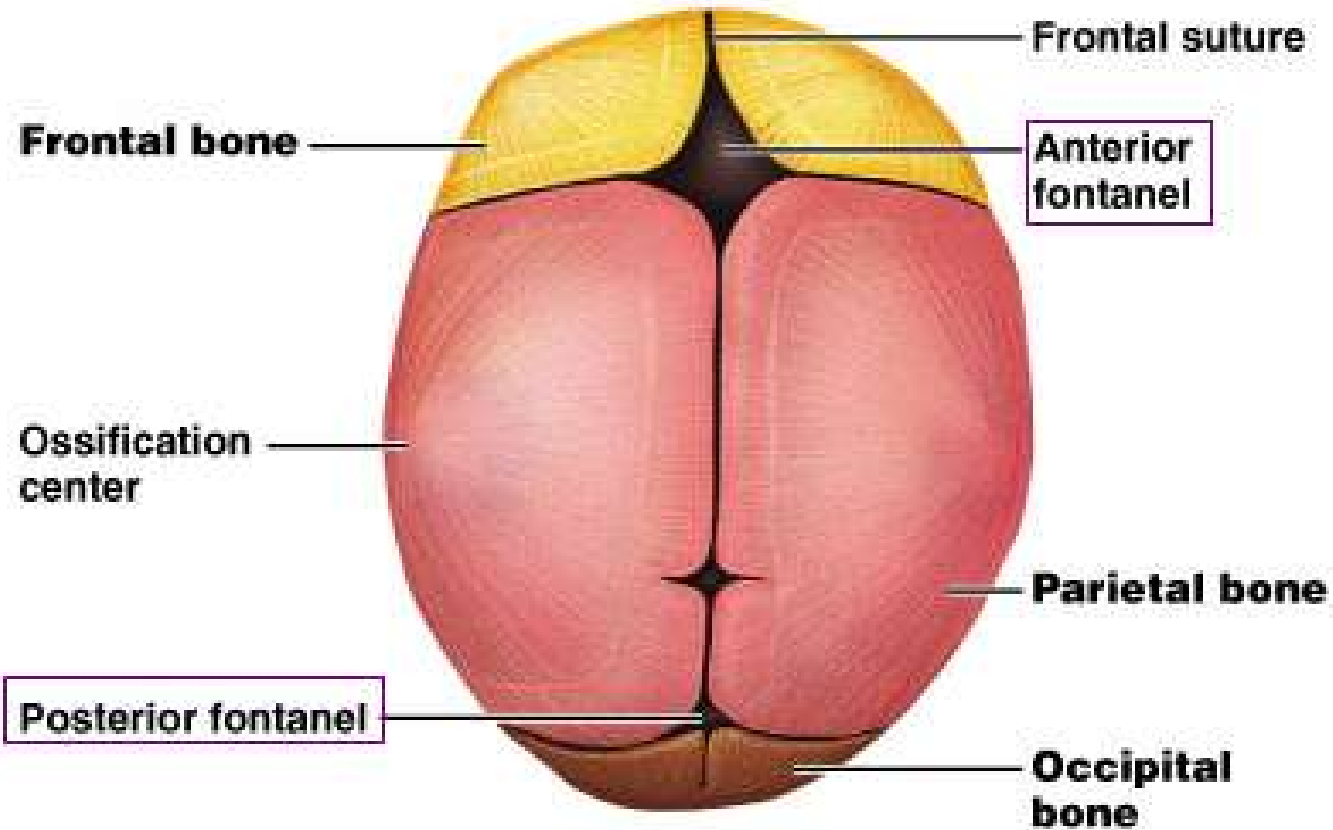


FONTANELLES IN NEWBORN

- Fontanelles are soft spots on a baby's head which, during birth, enable the bony plates of the skull to flex, allowing the child's head to pass through the birth canal.
- The ossification of the bones of the skull cause the fontanelles to close over by a child's second birthday.
- The closures form the sutures of the neurocranium.

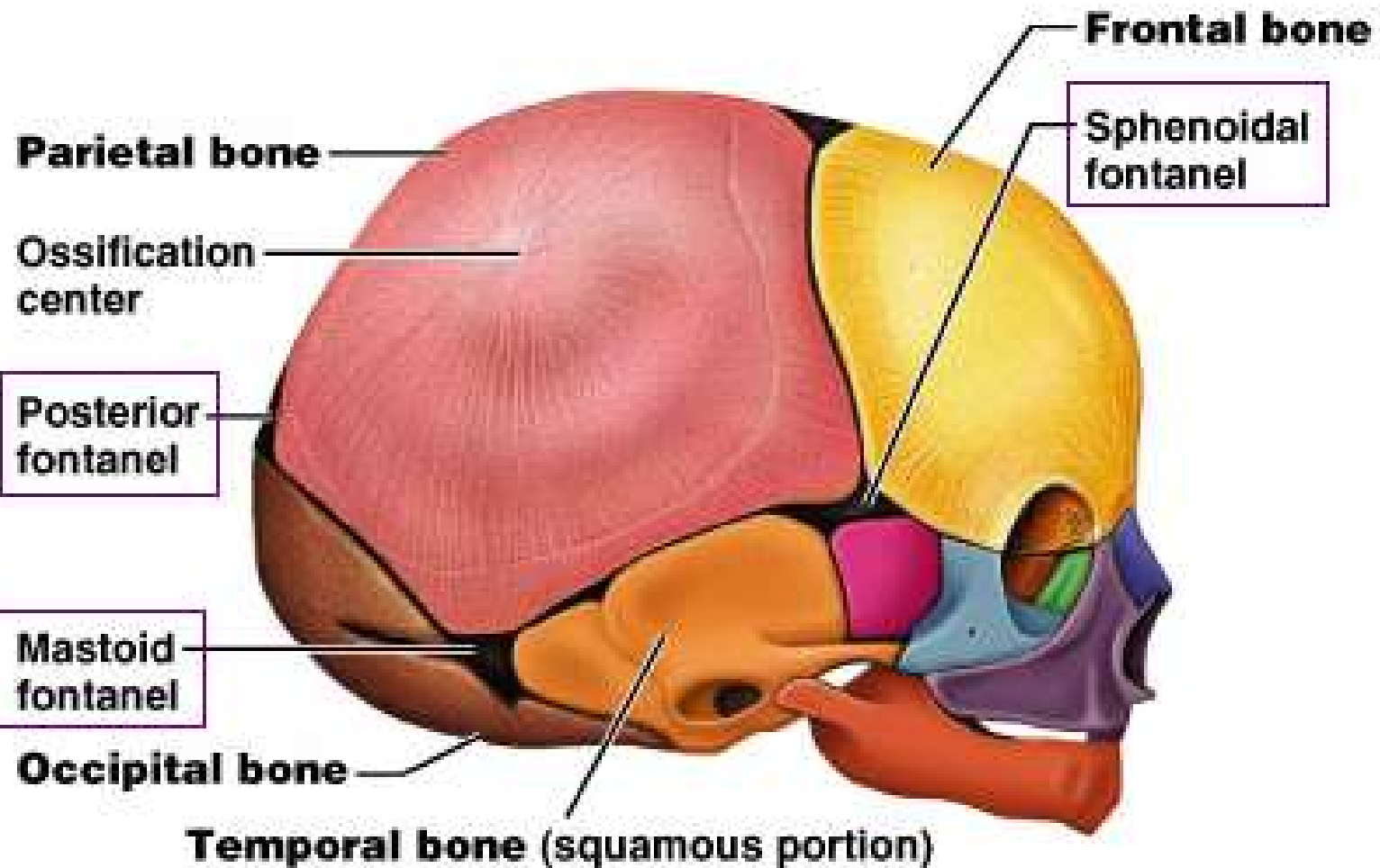


FONTANELLES



(a) Superior view

FONTANELLES



(b) Lateral view

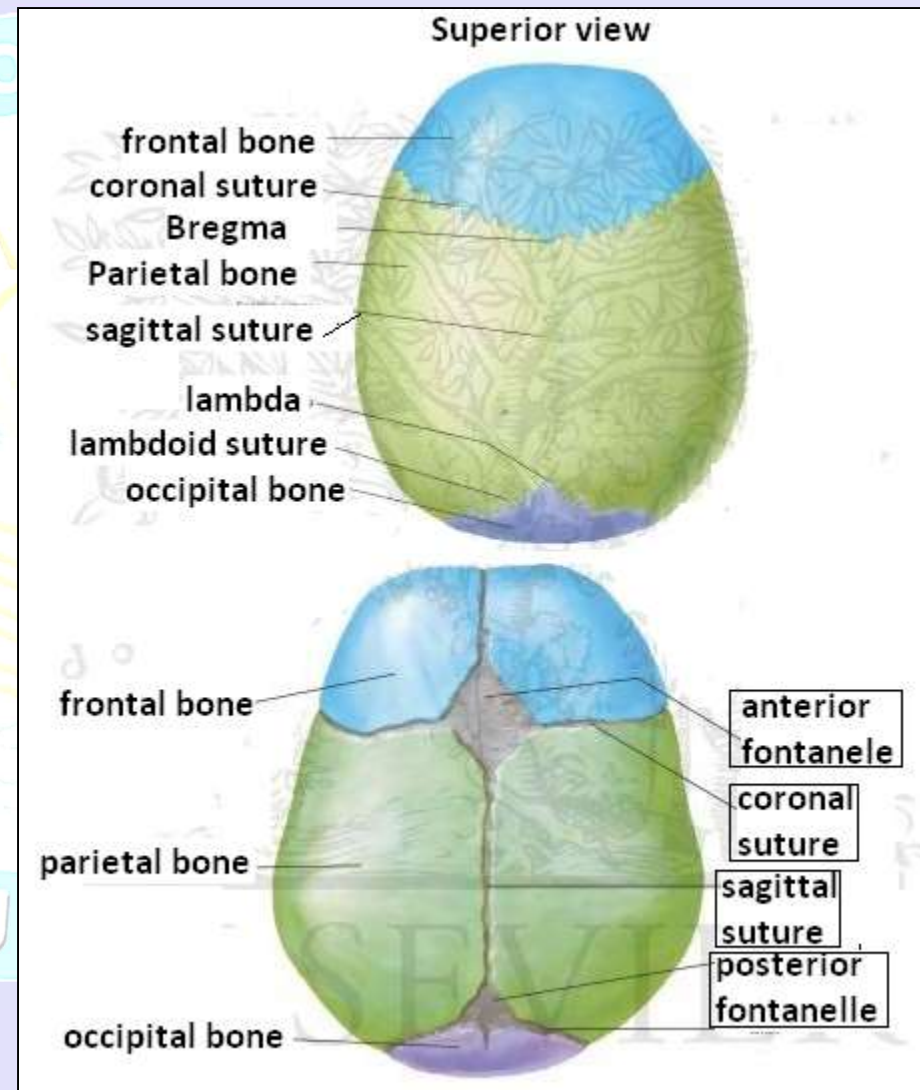
STUDY OF SKULL

Skull can be studied from different views. The views so obtained are termed the **normae** of the skull

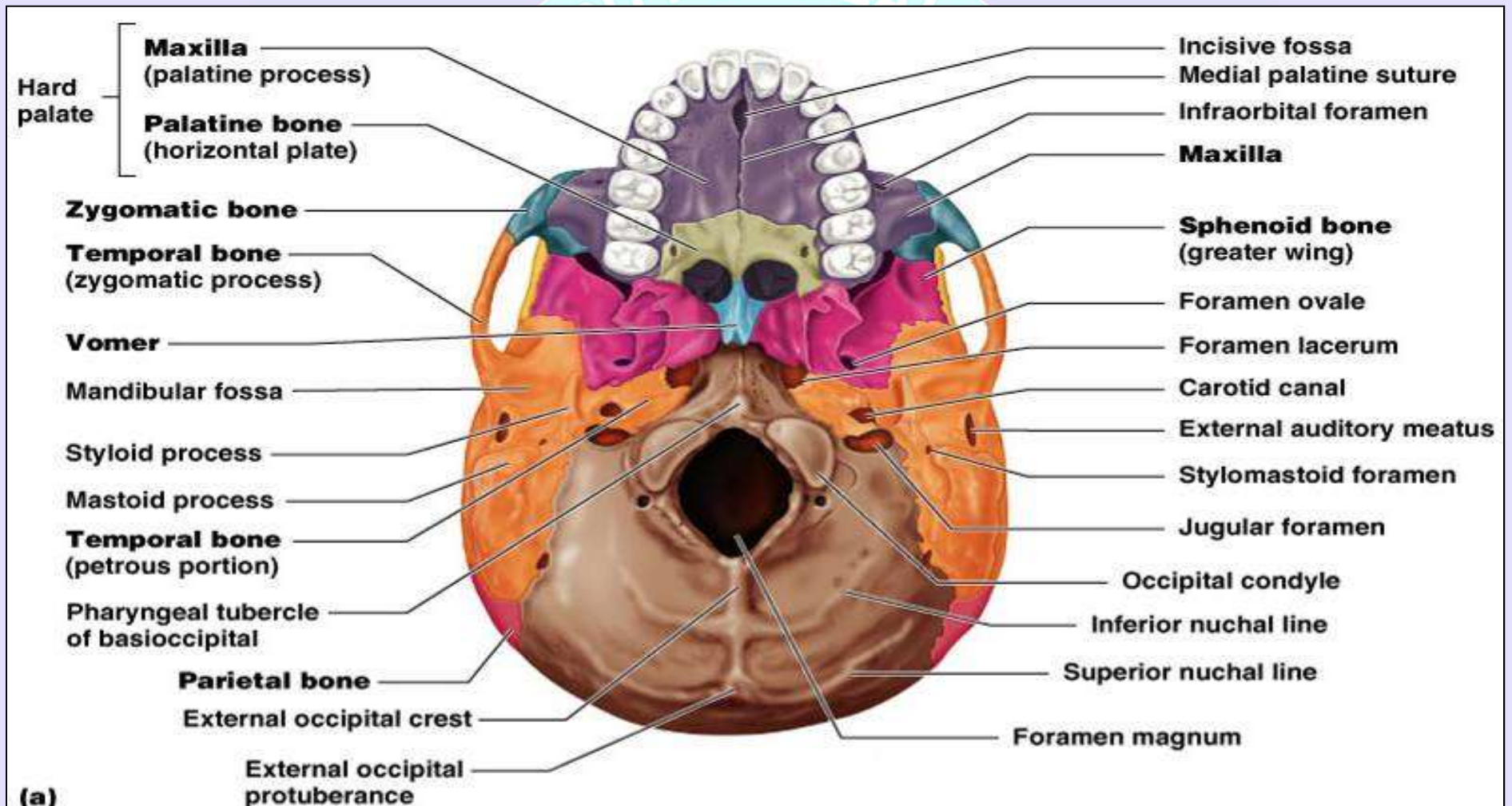
- From Above- Norma Verticalis
- From Below- Norma Basalis
- From Front- Norma Frontalis
- From Back- Norma Occipitalis
- From Side- Norma Lateralis
- From Inside – Interior of skull

SKULL SUPERIOR VIEW (NORMA VERTICALIS)

- Four sutures mark the articulations of the parietal bones
 - Coronal suture – articulation between parietal bones and frontal bone anteriorly
 - Sagittal suture – where right and left parietal bones meet superiorly
 - Lambdoid suture – where parietal bones meet the occipital bone posteriorly
 - Squamosal or squamous suture – where parietal and temporal bones meet



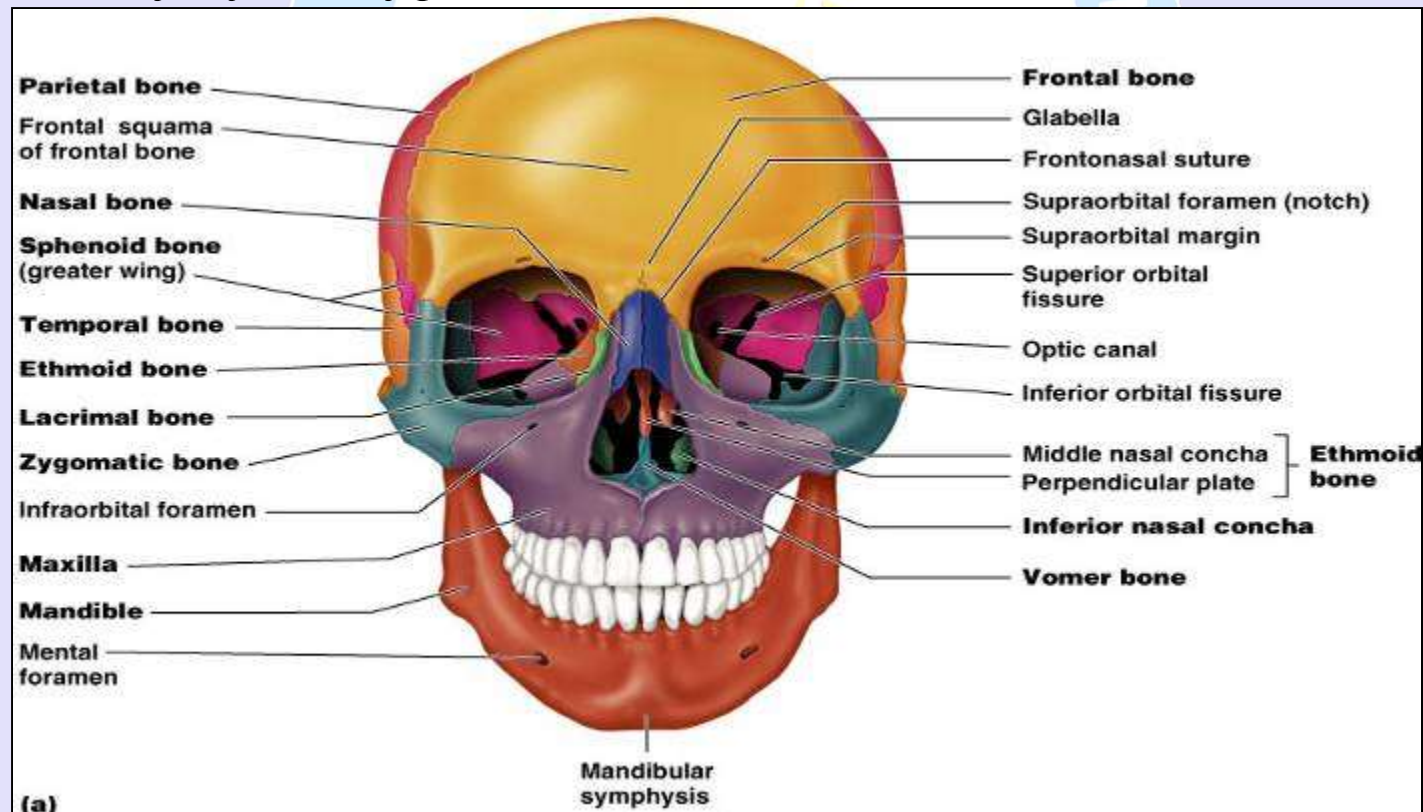
INFERIOR VIEW OF THE SKULL (NORMA BASALIS)



THE SKULL

ANTERIOR VIEW (NORMA FRONTALIS)

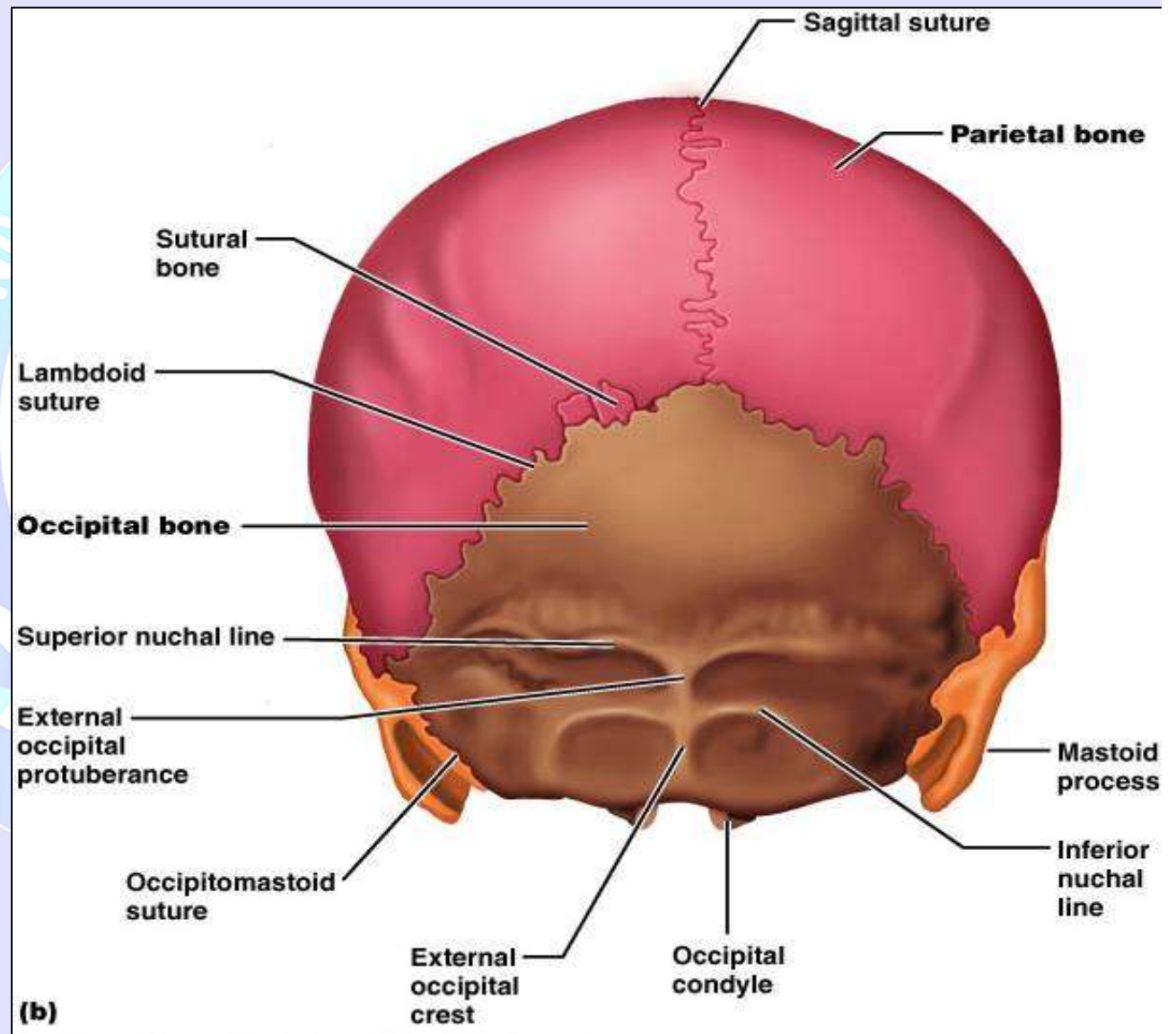
- oval outline
- Limited
 - *above* by the frontal bone
 - *below* by the body of the mandible
 - *laterally* by the zygomatic bones and the mandibular rami.



SKULL POSTERIOR VIEW (NORMA OCCIPITALIS)



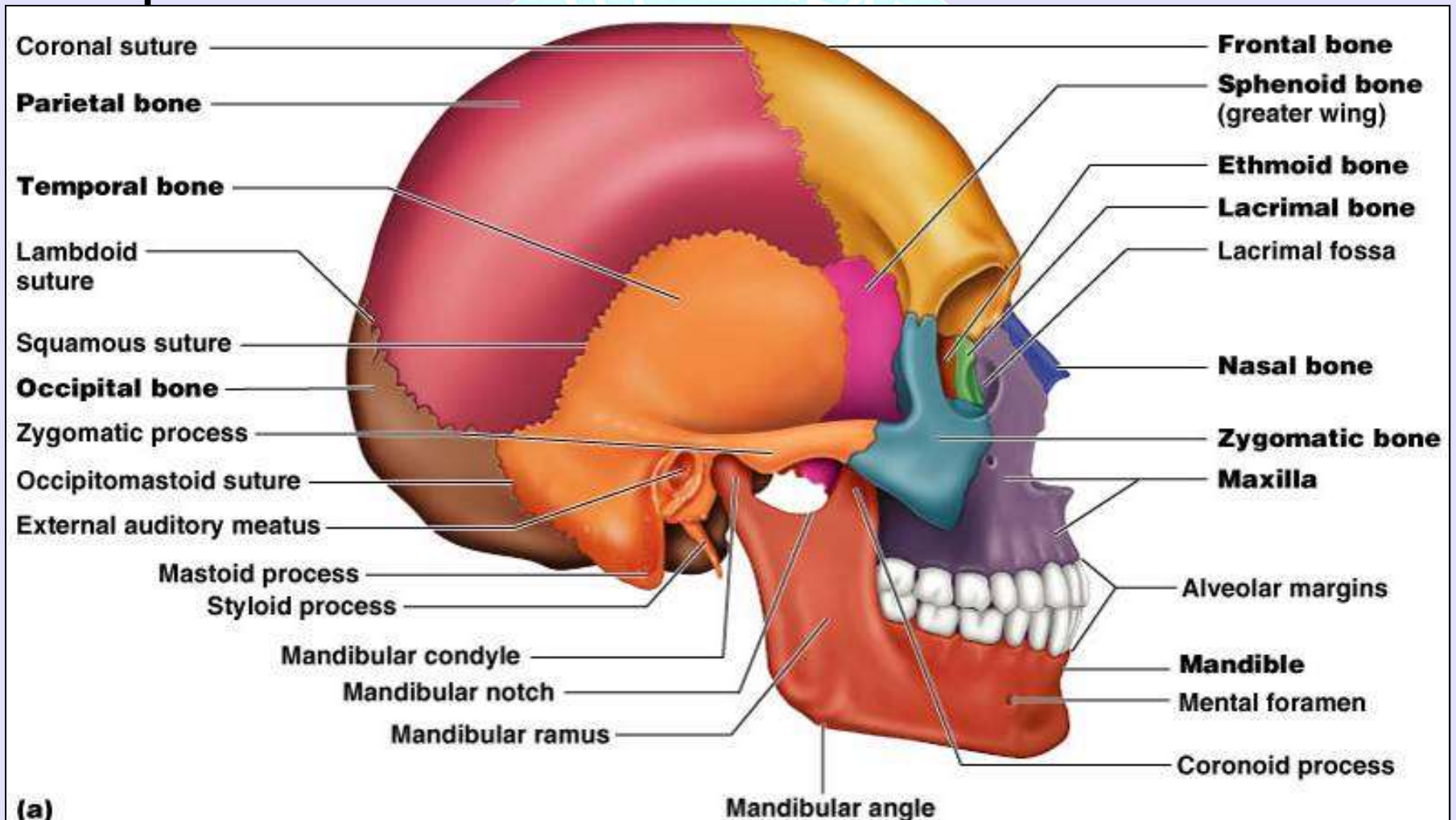
- Circular outline.
- Forms most of skull's posterior and base
- Major markings include
 - **Sagittal suture**
 - **Lambdoid suture**
 - **external occipital protuberance**
 - **mastoid foramen**
 - **foramen magnum,**
 - **occipital condyles**
 - **hypoglossal canal**



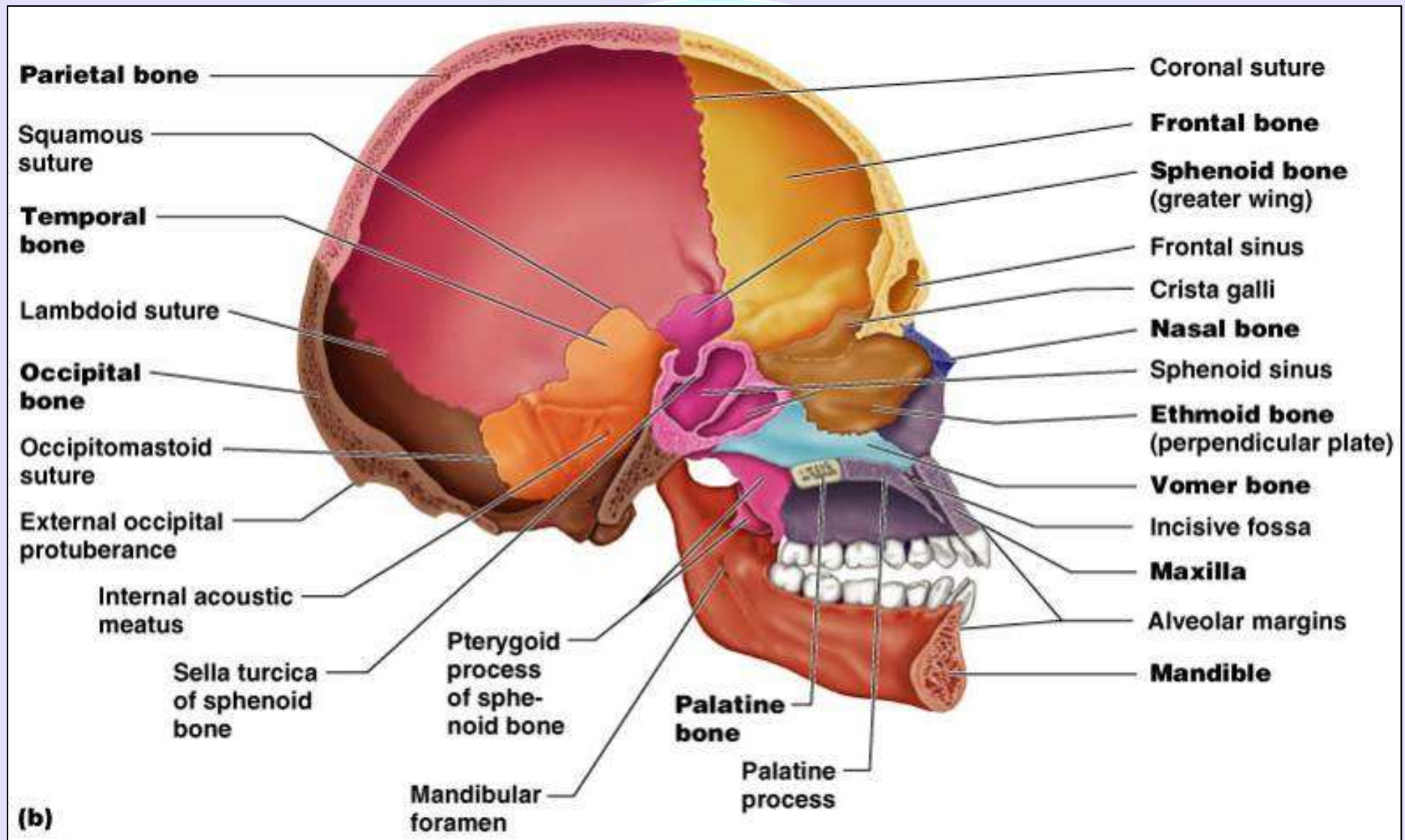
Skull: Lateral View (Norma Lateralis)



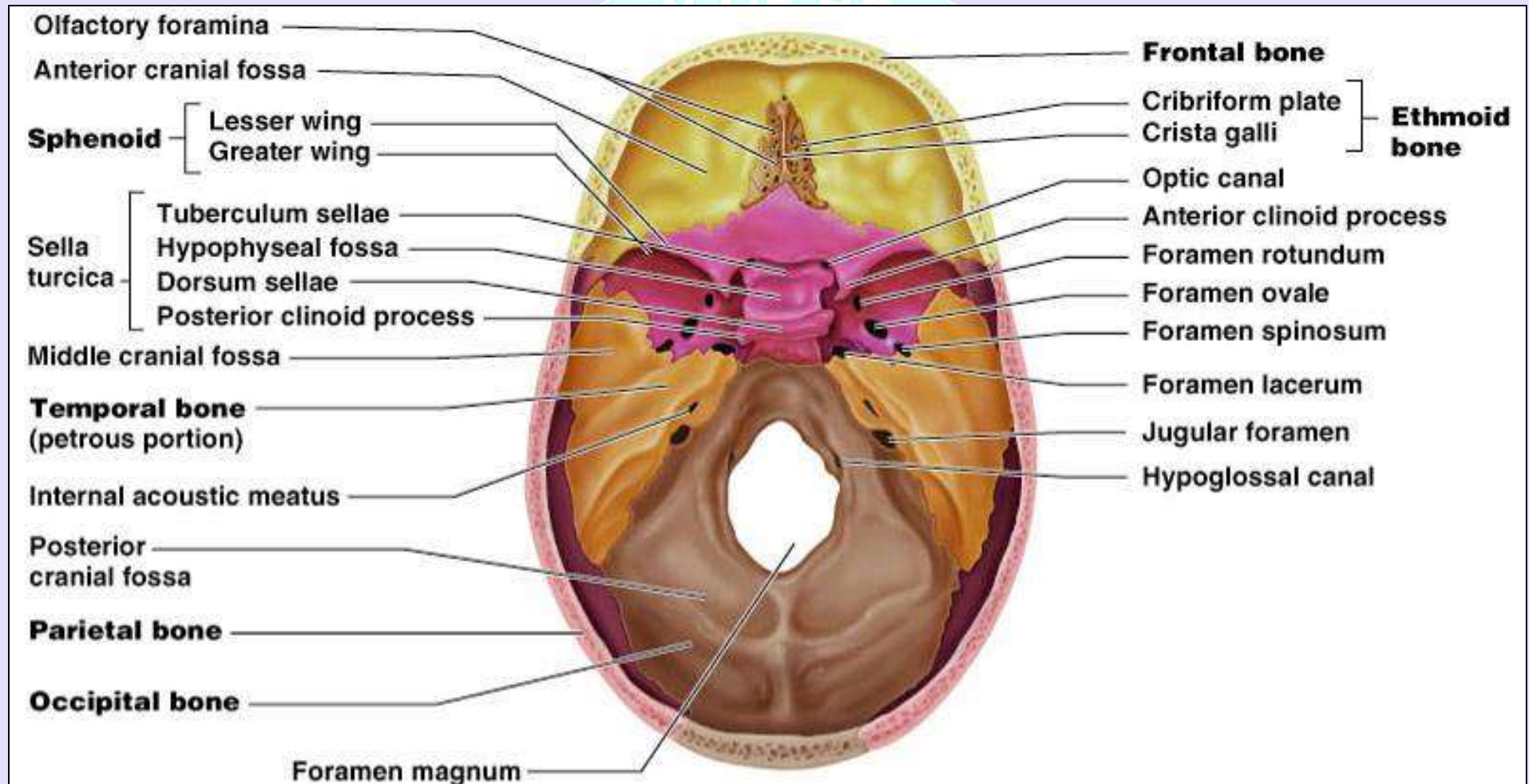
- Form most of the superior and lateral aspects of the skull



MIDSAGITTAL LATERAL ASPECTS OF THE SKULL



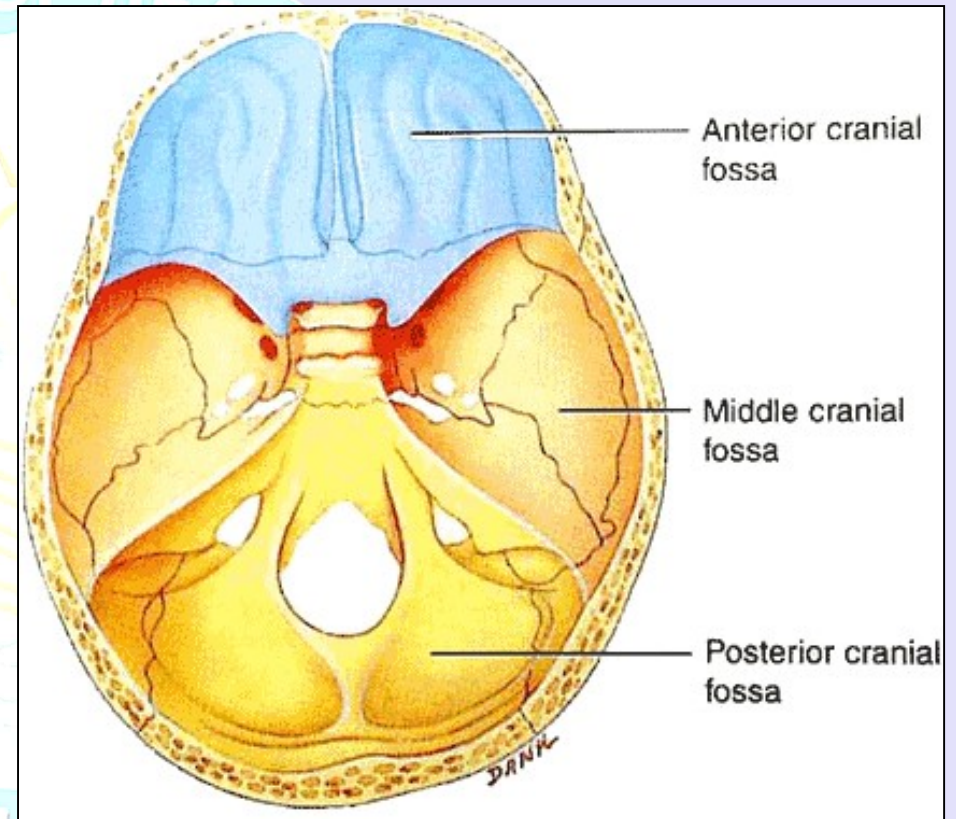
INTERIOR OF THE SKULL (CRANIAL FOSSA)



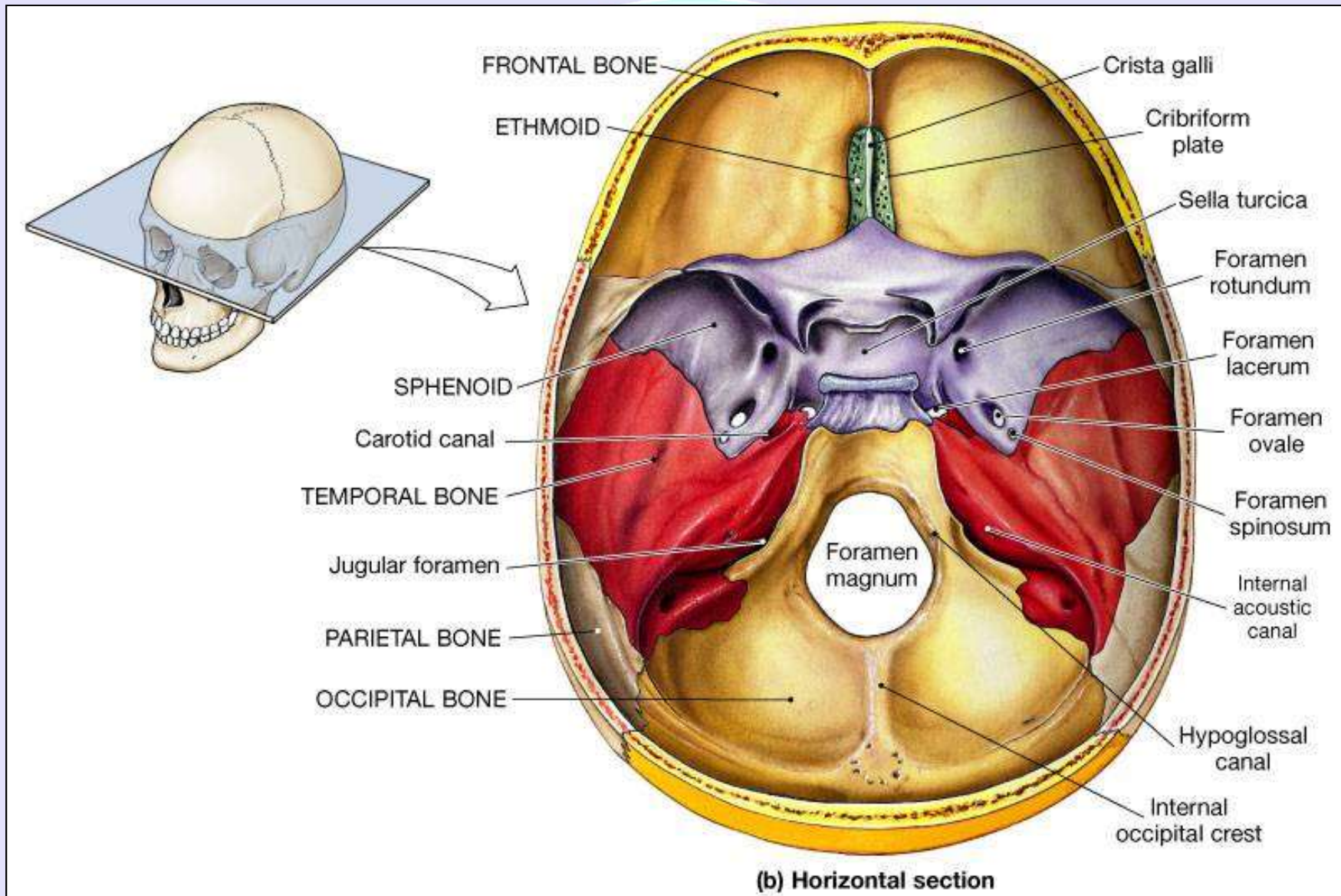
THE CRANIAL FOSSAE

Cranial fossa – curving depression of the cranial floor

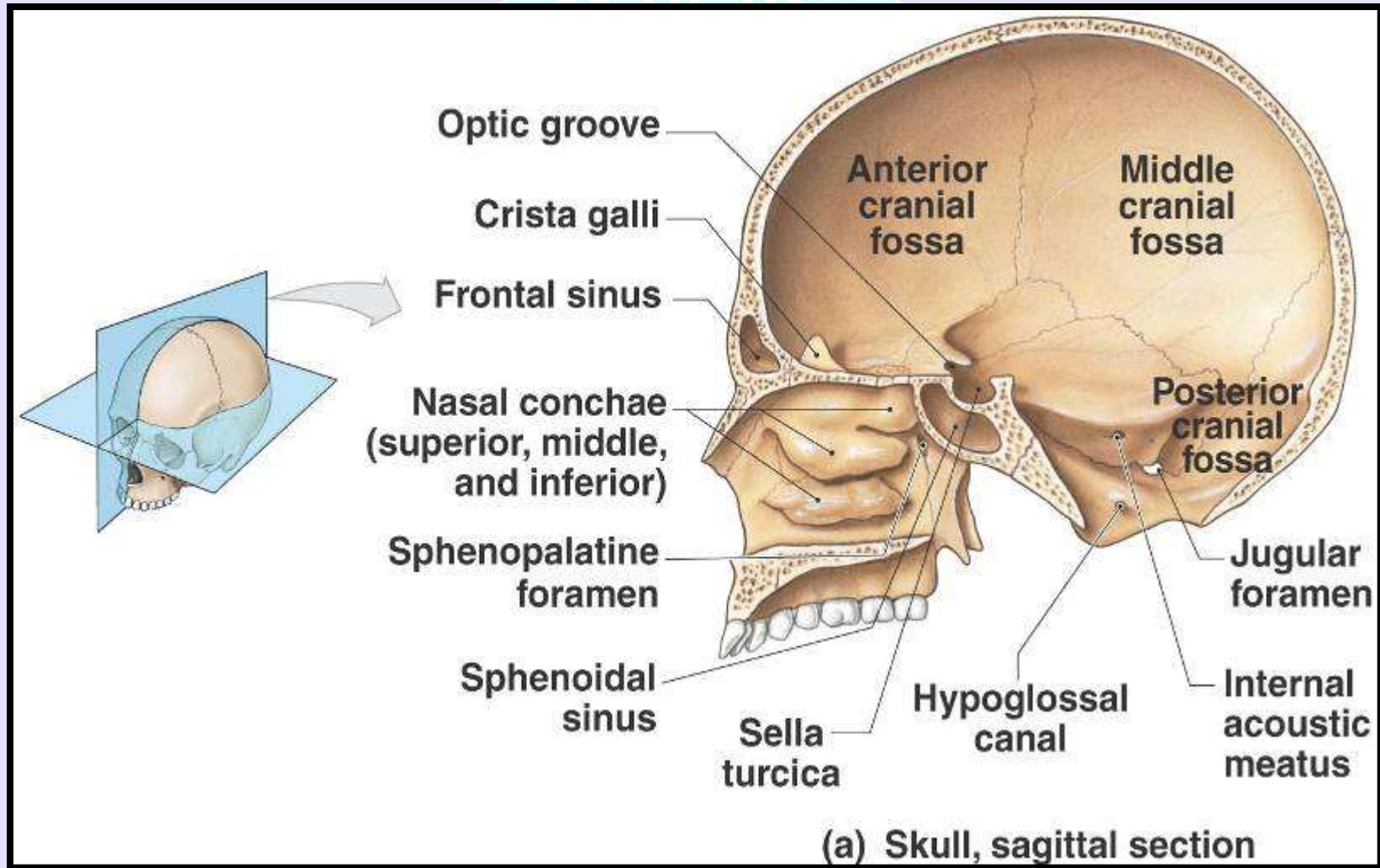
- Anterior cranial fossa formed by:
frontal bone, ethmoid, lesser wing of the sphenoid;
cradles the frontal lobes of the cerebral hemispheres
- Middle cranial fossa is formed by:
sphenoid, temporal, parietal bones; cradles the temporal lobes of the cerebral hemispheres, the diencephalon, and mesencephalon
- Posterior cranial fossa is formed primarily by:
occipital bone, with contributions from the temporal and parietal bones
 - supports the occipital lobes of the cerebral hemispheres, the cerebellum, and the pons and medulla oblongata (brain stem)



THE SECTIONAL ANATOMY OF THE SKULL



THE CRANIAL FOSSAE



THE CRANIAL FOSSAE

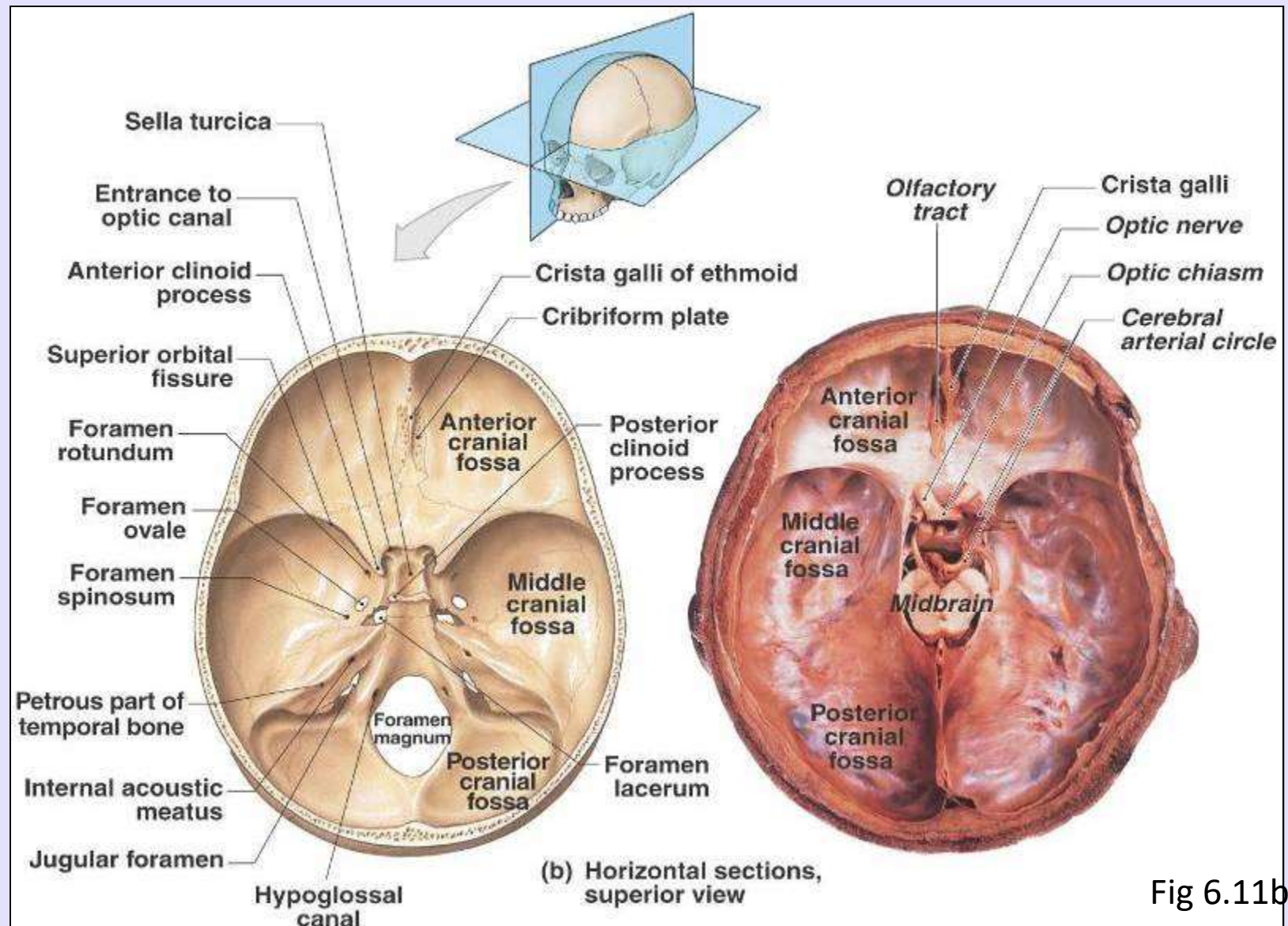
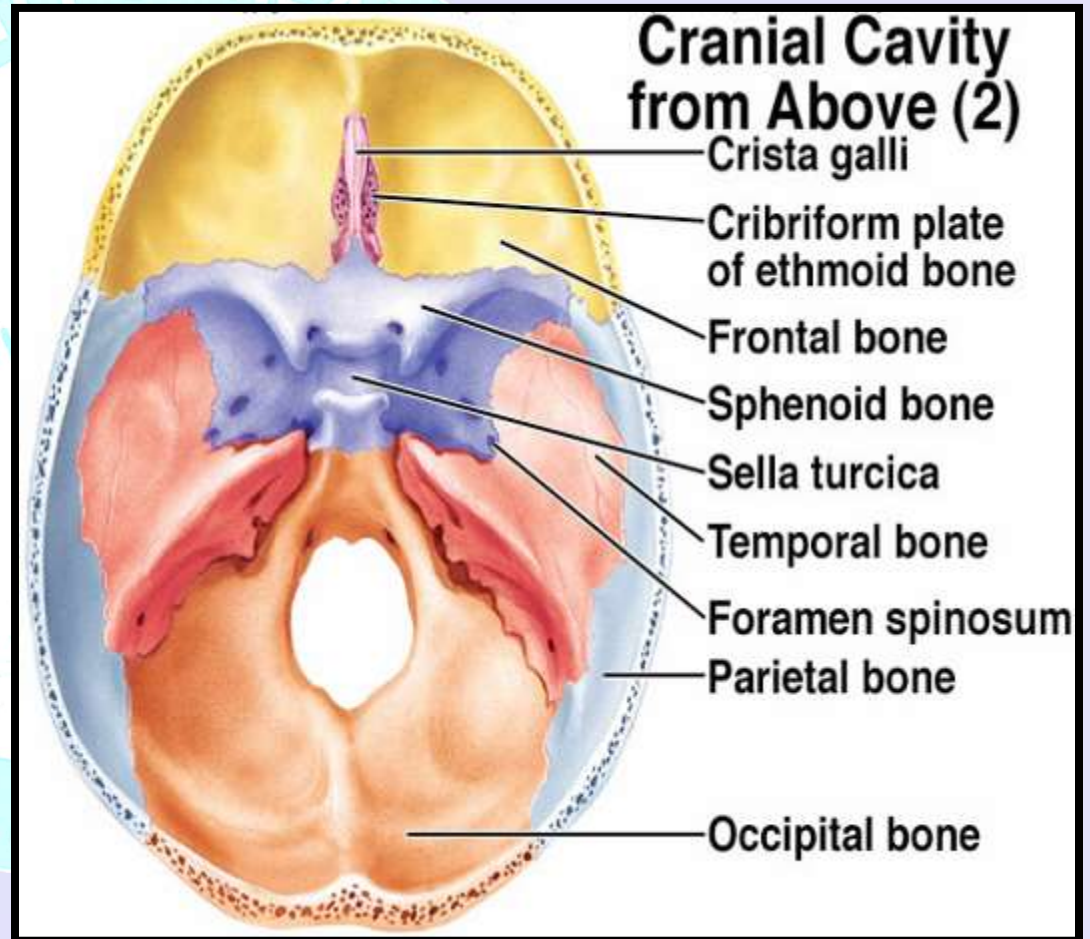


Fig 6.11b

INTERIOR OF THE CRANIAL CAVITY

- **Cranial cavity:**
occupied by the brain
- **Calvaria (skull cap):**
upper dome-like portion of skull
- Floor divided into anterior, middle, and posterior fossae
- **Crista galli:**
prominent ridge in center of anterior fossa. Point of attachment for the **dura mater** (one of the **meninges**)

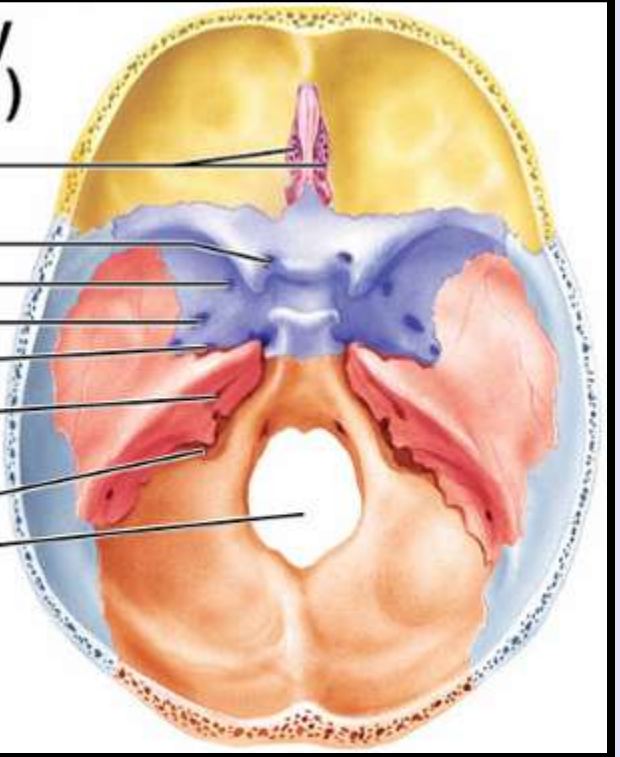


INTERIOR OF THE CRANIAL CAVITY

- **Olfactory fossae** lateral to crista galli. Olfactory bulb within
 - **Cribriform plate of the ethmoid forms** floor of olfactory fossae
 - Olfactory nerves pass through the foramina of the cribriform plate
- **Sella turcica**: part of sphenoid bone that houses the **pituitary gland**
- **Foramen magnum**: opening where brain attaches to spinal cord

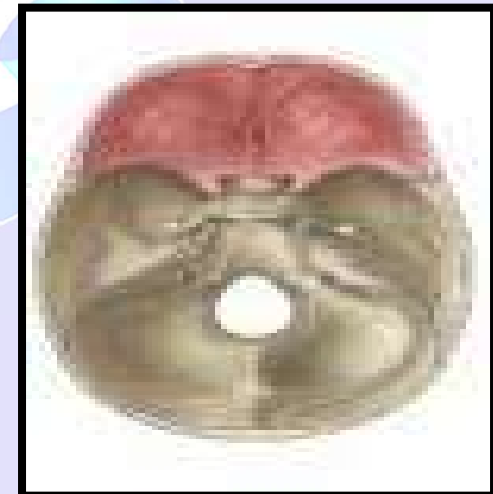
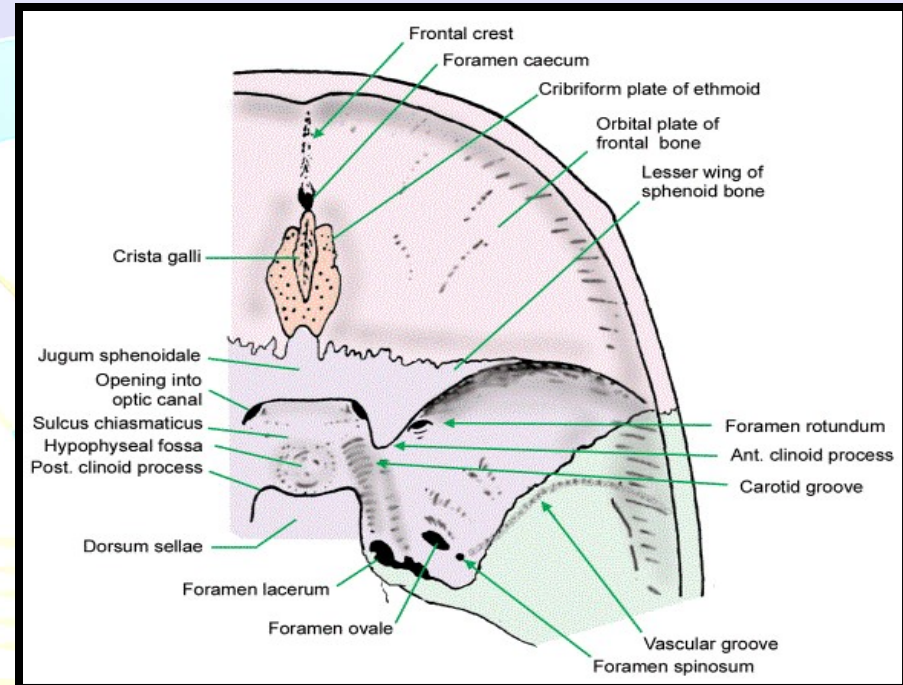
Cranial Cavity from Above (1)

Olfactory foramina
Optic canal
Foramen rotundum
Foramen ovale
Foramen lacerum
Internal acoustic meatus
Jugular foramen
Foramen magnum



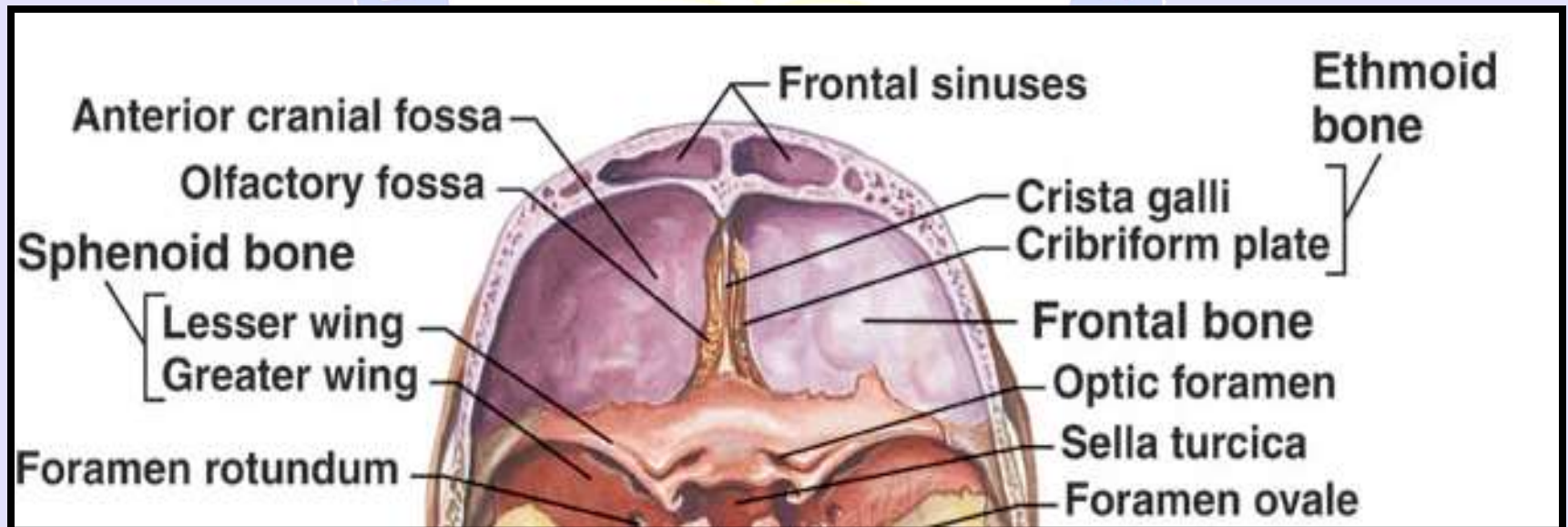
ANTERIOR CRANIAL FOSSA

- The floor of the anterior fossa is formed by:
 - Orbital plates of the frontal,
 - Cribriform plate of the ethmoid
 - small wings and front part of the body of the sphenoid
- It is limited behind by the posterior borders of the small wings of the sphenoid and by the anterior margin of the chiasmatic groove.
- It is traversed by the **frontoethmoidal suture**
sphenoethmoidal suture
sphenofrontal sutures.



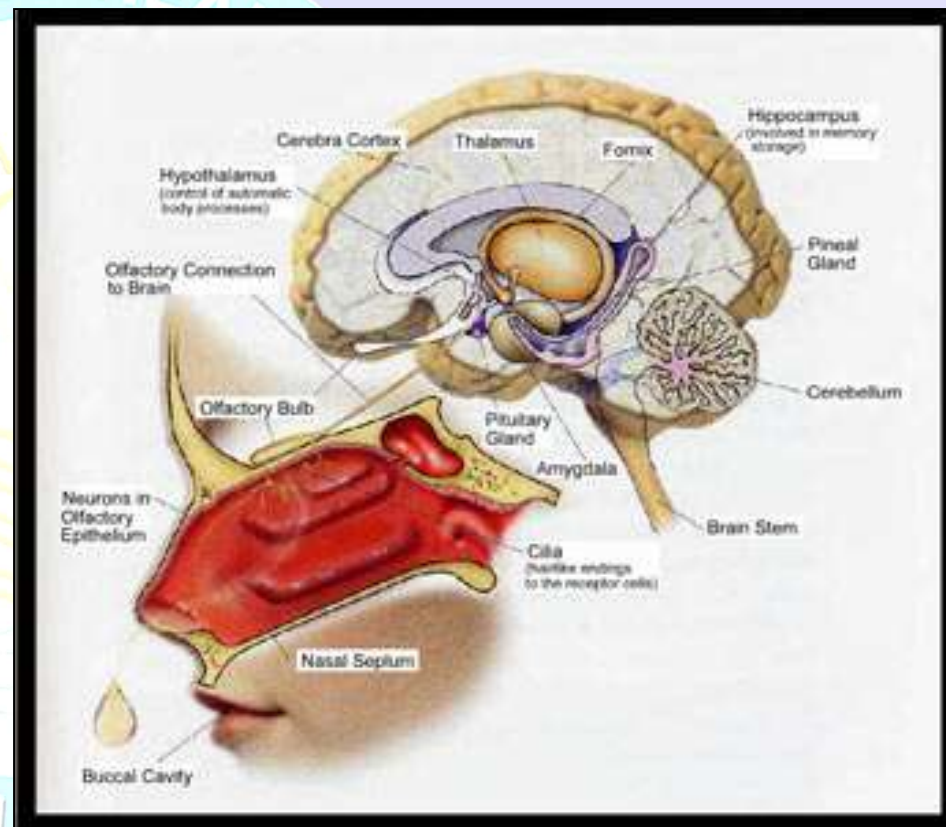
ANTERIOR CRANIAL FOSSA

- The central portion corresponds with the roof of the nasal cavity, and is markedly depressed on either side of the crista galli.
- It presents, in and near the median line, from before backward, the commencement of the **frontal crest** for the attachment of the falx cerebri
- **Foramen cecum,**
 - The frontal crest of the frontal bone ends below in a small notch which is converted into a foramen, by articulation with the ethmoid
 - It transmits a vein from the nose to the superior sagittal sinus



ANTERIOR CRANIAL FOSSA

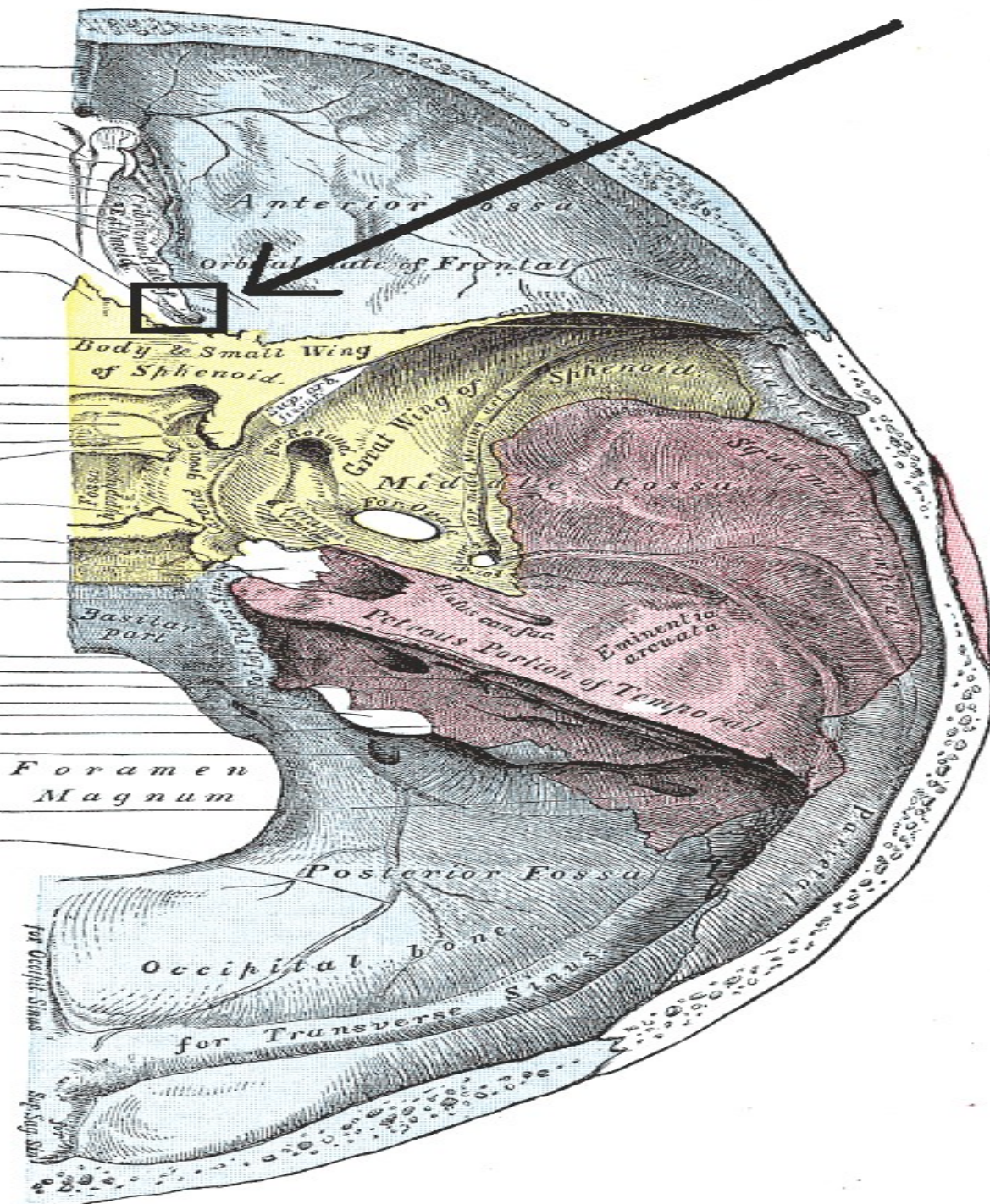
- **Crista galli**, ridge behind the foramen cecum, the free margin of which affords attachment to the falx cerebri
- **Olfactory groove** on either side of the crista galli, formed by the cribriform plate,
- Supports the olfactory bulb and presents foramina for the transmission of the olfactory nerves,
- In front a slit-like opening for the nasociliary nerve.



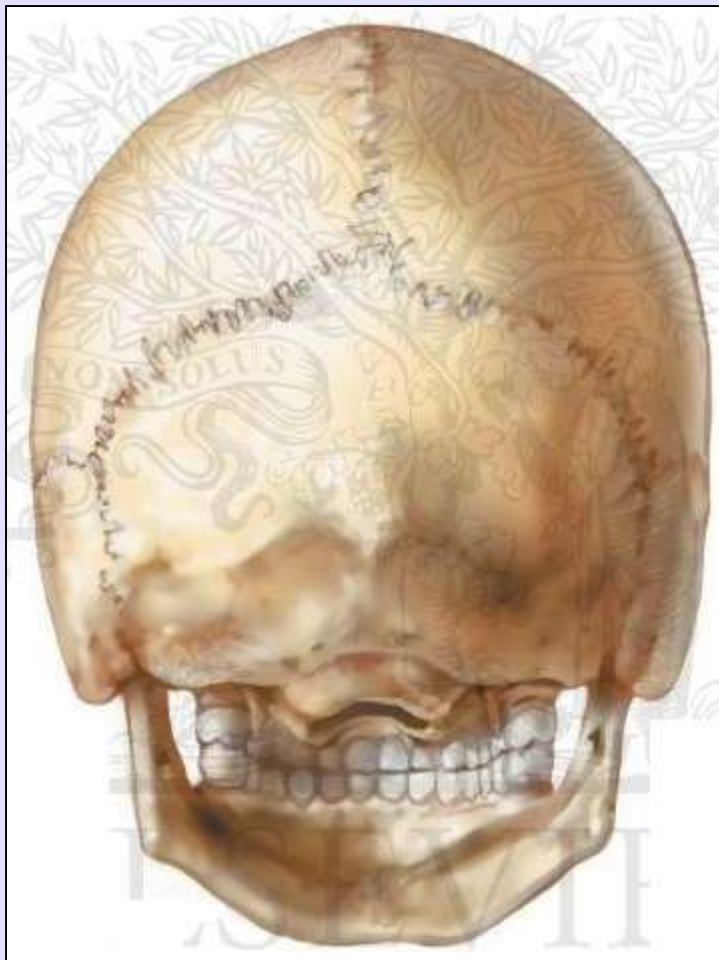
ANTERIOR CRANIAL FOSSA

- **Anterior ethmoidal foramen**
 - situated about the middle of the lateral margin of the olfactory groove,
 - Transmits the anterior ethmoidal vessels and the anterior ethmoidal nerve
- **Posterior ethmoidal foramen**
 - Opens at the back part of this margin under cover of the projecting lamina of the sphenoid.
 - transmits the posterior ethmoidal vessels and nerve

Groove for super. sagittal sinus
 Grooves for anter. meningeal vessels
 Foramen cæcum
 Crista galli
 Slit for nasociliary nerve
 Groove for nasociliary nerve
 Anterior ethmoidal foramen
 Orifices for olfactory nerves
 Posterior ethmoidal foramen
 Ethmoidal spine
 Olfactory grooves
 Optic foramen
 Chiasmatic groove
 Tuberculum sellae
 Anterior clinoid process
 Middle clinoid process
 Posterior clinoid process
 Groove for abducent nerve
 Foramen lacerum
 Orifice of carotid canal
 Depression for semilunar ganglion
 Internal acoustic meatus
 Slit for dura mater
 Groove for superior petrosal sinus
 Jugular foramen
 Hypoglossal canal
 Aquæductus vestibuli
 Condylloid foramen
 Mastoid foramen
 Posterior meningeal grooves



Name the Norma??

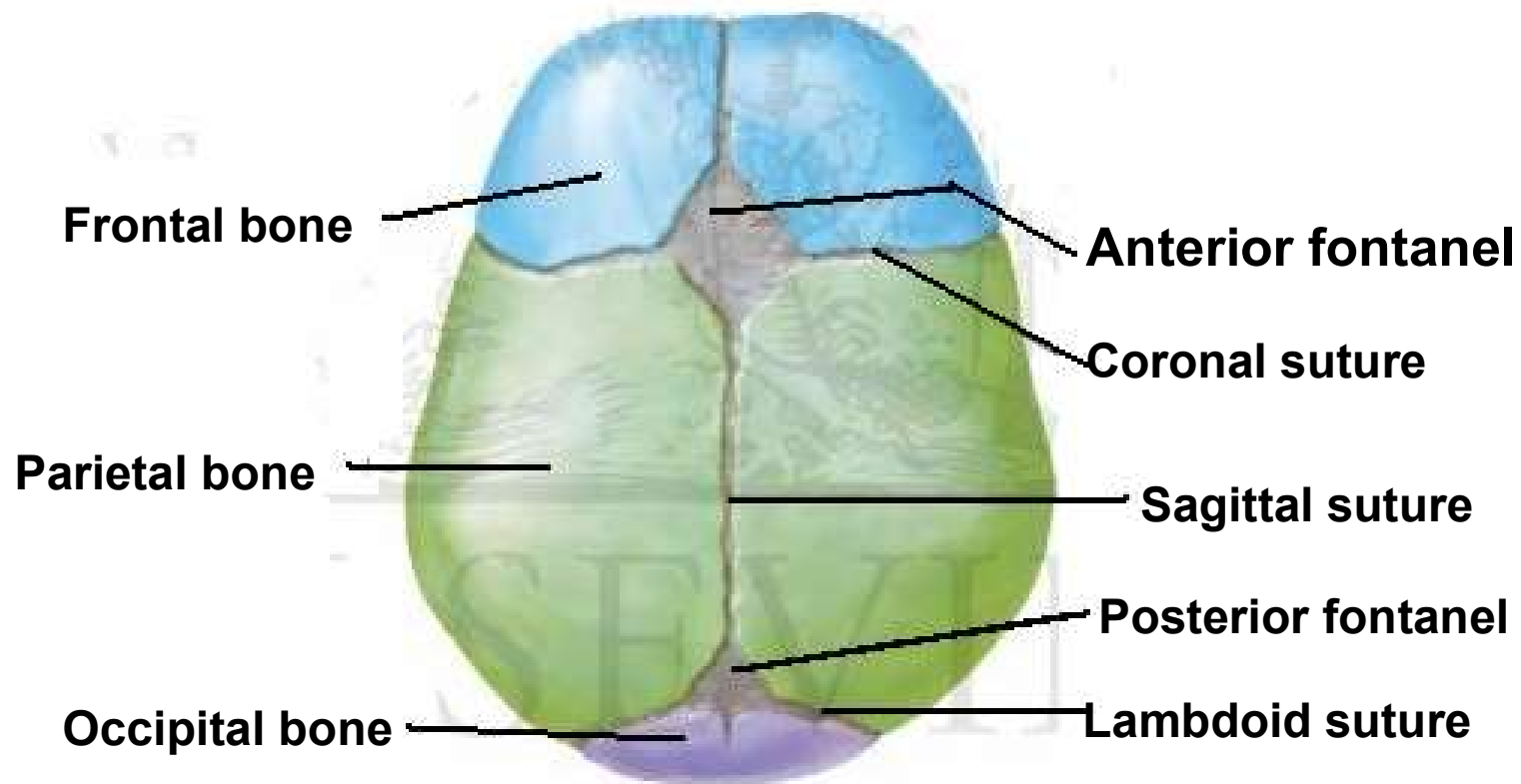


Norma occipitalis



Norma basalis

Identify??



REFERENCES

- KLM clinical anatomy correlation.
- Snell's Neuro Anatomy .
- RJ Last's of Anatomy.

